

Language $\rightarrow$ communication tool.
Program $\rightarrow$ set of instruction.

} used to communicate with computers to perform set of instruction.

Types

Low level.

- 1) Assembly Language
- 2) Machine Language.
(0-1)

high level.

- 1) Procedural oriented.
C

- 2) Object oriented
Python

- 3) function oriented. (JavaScript)

Compiler/Interpreter

Java is Partially Object oriented.

Translates high level Programming Language
into machine Language.
Compiler $\rightarrow$ Interpreter

* Translates the entire Program.

* Translates Line by Line.

* Time Take is Less

* Time Taken is more.

* Difficult to debug

* Easy to debug.

* C, C++

* Interpreter.

Java

-> it was developed by James Gosling at Sun micro systems.

Structure of java

Documentation Section

single line comment
//

multi line comment.
/* _____ */

Class Initialization

main()

— Public Static

— class Main ^{file name}
void main(String args[]) {
}

Static method will access only

Static variable / method

output statement

-> System.out.print

-> System.out.println

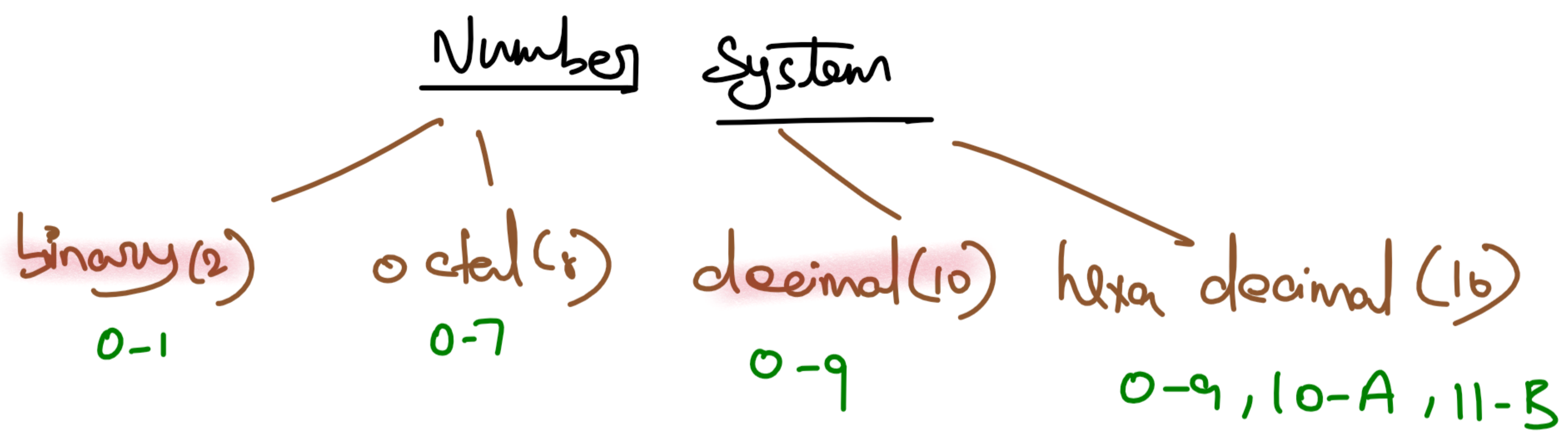
} 1st Print then if it is a 'ln' it will create a new line.

```
System.out.print("Nanthish");  
System.out.print("Samantha");  
System.out.println("Nanthish");  
System.out.print("Samantha");  
System.out.println("Nanthish");  
System.out.print("Samantha");
```

Escape sequence

ln - new line

lt - tab space.



Decimal to any base

2	33
2	16 - 1
2	8 - 0
2	4 - 0
2	2 - 0
2	1 - 0
2	0 - 1

$\Rightarrow (100001)$

2^5	2^4	2^3	2^2	2^1	2^0
32	16	8	4	2	1
1	0	0	0	0	1

10 $\Rightarrow$ 1 0 1 0

20 $\Rightarrow$ 1 0 1 0 0

32	16	8	4	2	1
1	0	1	0	1	0

$\Rightarrow 42$

6 bits representation

neg representation

2^5	2^4	2^3	2^2	2^1	2^0 (LSB)
32	16	8	4	2	1

$[-32, 31]$
 $[-2^5, 2^5-1]$

4 bits

2^3	2^2	2^1	2^0
8	4	2	1

$[-8, 7]$
 $[-2^3, 2^3-1]$

Data Types

collection of raw facts

byte — 1 byte = 8 bits $[-2^7, 2^7-1] \Rightarrow [-128, 127]$
short — 2 bytes = 16 bits $[-2^{15}, 2^{15}-1]$
int — 4 bytes = 32 bits $[-2^{31}, 2^{31}-1]$
long — 8 bytes = 64 bits $[-2^{63}, 2^{63}-1]$

} — whole Number.

Point Values

float — 4 bytes — 32 bits $[-2^{31}, 2^{31}-1]$
double — 8 bytes — 64 bits $[-2^{63}, 2^{63}-1]$
char — 2 bytes — 16 bits $[-2^{15}, 2^{15}-1]$
boolean — 1 bit [true / false]

Primitive data type

byte, short, int, long,
float, double, char,
boolean

Non-Primitive data type

Array Array List

Variable

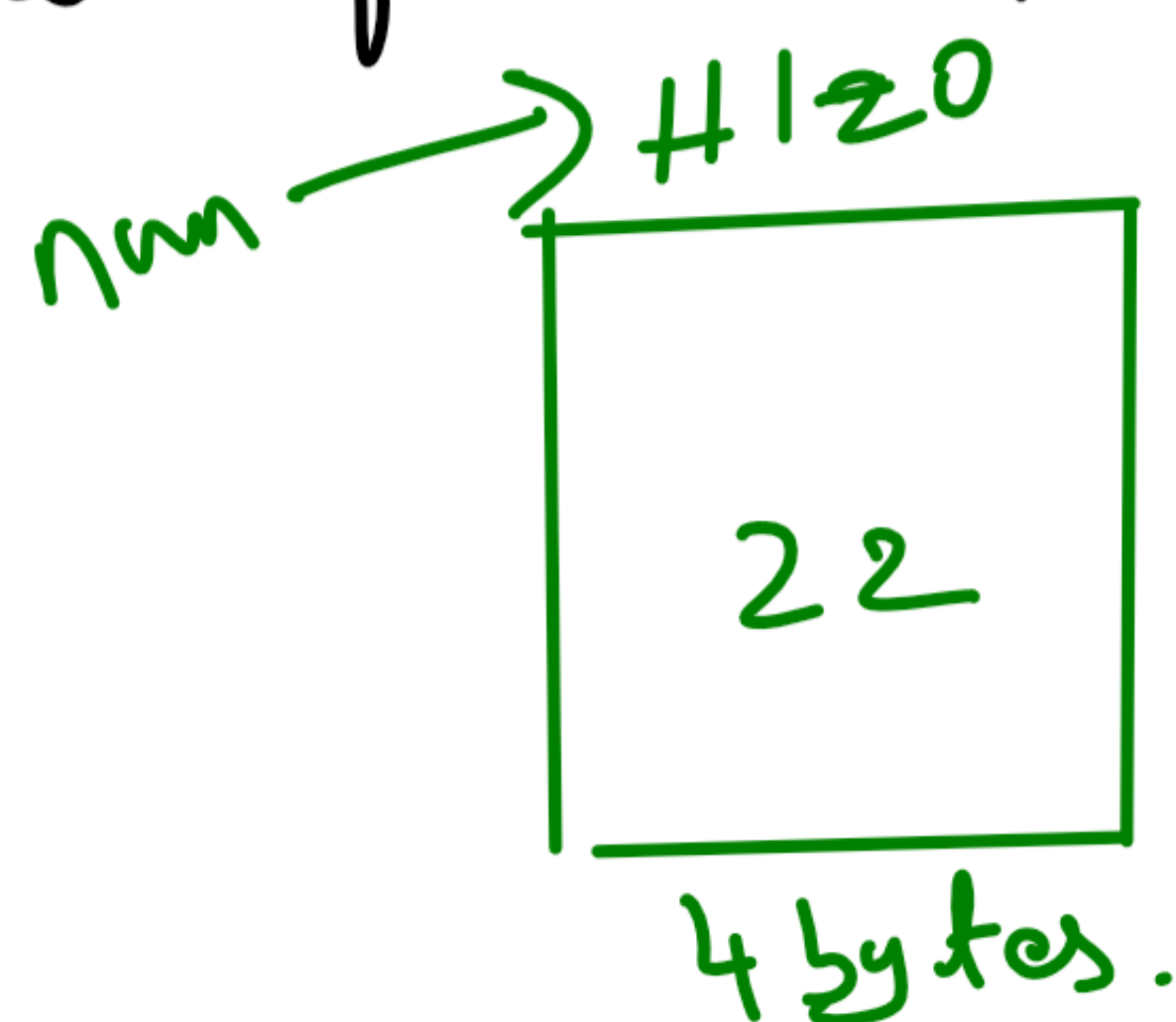
used to store the address of data.

Syntax

data-type var-name;

int num;

num = 22;



Rules of Variables (s, s.D, k)

- 1) Space is not allowed.
- 2) Special symbol is not allowed Except (_, \$)
- 3) Should not start with Digits.
- 4) Variable name can not be a keyword
Reserved words.

Value initializing to Variable.

1) Compile Time / static initialization

```
public class Main{  
    public static void main(String args[]){  
        int num=22;  
        System.out.println(num);  
    }  
}
```

2) Run Time / dynamic Initialization (getting value from user).

1) import Scanner class

2) Scanner (Obj-name) = new Scanner(System.in);
You can get the value from user.

int

```
Scanner scn=new Scanner(System.in);  
int num=scn.nextInt();  
System.out.println(num);
```



```

import java.util.*;
public class Main{
    public static void main(String args[]){
        Scanner scn=new Scanner(System.in);
        //String val=scn.next(); it reads one word
        //String val=scn.nextLine();
        //int num=scn.nextInt();
        //float val=scn.nextFloat();
        //double val=scn.nextDouble();
        //char val=scn.next().charAt(0);
        System.out.println(val);
    }
}

```

Operators :- it is a symbol which is used to perform operation between any 2 operands.

$a=10, b=5$
 $\boxed{c=a+b}$

1) Arithmetic Operator (+, -, *, /, %)

$$10/5 = 2$$

$$8/3 = 2$$

$$10/2 = 5$$

$$10\%5 = 0$$

$$8\%3 = 2$$

$$10\%2 = 0$$

$$7/9$$

$$7\%9$$

$$\begin{array}{r}
 5 - 1 \\
 2 \overline{) 10} \\
 \underline{10} \\
 0 = \%
 \end{array}$$

$$\begin{array}{c}
 UV \quad LV \\
 \swarrow \quad \searrow \\
 0 \quad UV
 \end{array}$$

$$\begin{array}{l|l}
 9/7 = 1 & 7/9 = 0 \\
 9\%7 = 2 & 7\%9 = 7
 \end{array}$$

Functions → set of instruction to perform specific task.

Function

Pre defined

max(), sqrt()

User defined

1) with argument & with return type.

```
import java.util.*;
public class Main{
    public static int sum(int num1,int num2){
        return num1+num2;
    }
    public static void main(String args[]){
        Scanner scn=new Scanner(System.in);
        int a=scn.nextInt();
        int b=scn.nextInt();
        int res=sum(a,b); //function call
        System.out.println(res);
    }
}
```

2) with arguments & without return type.

```
import java.util.*;
public class Main{
    public static void sum(int num1,int num2){
        System.out.println(num1+num2);
    }
    public static void main(String args[]){
        Scanner scn=new Scanner(System.in);
        int a=scn.nextInt();
        int b=scn.nextInt();
        sum(a,b);
    }
}
```

2) Assignment Operator (=)

a = 10;
a = a + 5; a += 5;

a = 5
a = a * 5; a *= 5;

3) Relational Operator (>, ≥, <, ≤, ==, !=)

9 > 2 — True

9 < 20 — True

2 > 9 — False

20 < 9 — False

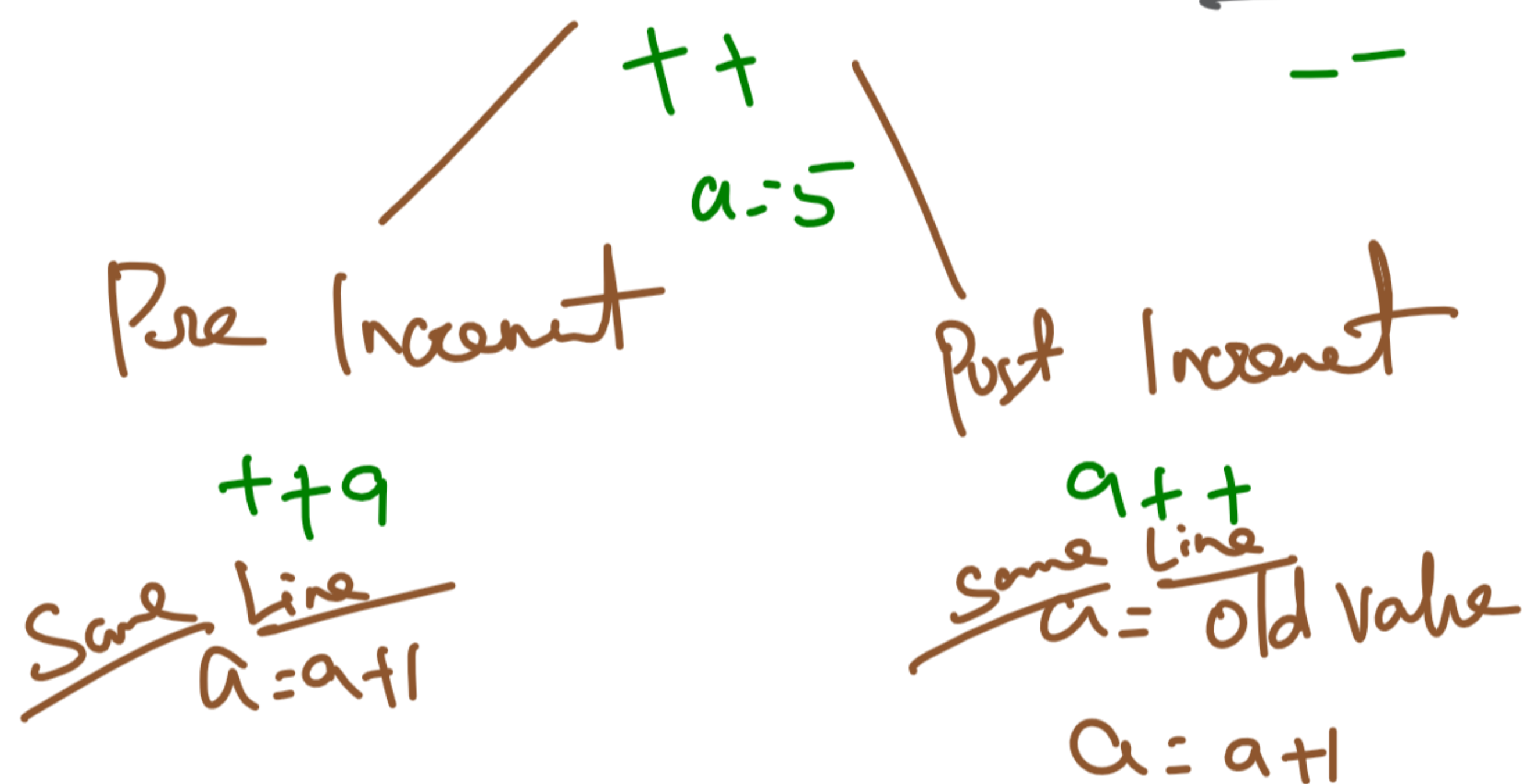
5 == 4 — F

5 != 9 — True.

4) Logical Operator (22, 11, !)

(5>3)		<u>22</u>	<u>11</u>		<u>NOT</u>
T	T	T	T		
T	F	F	T		T → F
F	T	F	T		F → T
F	F	F	F		
	(3>5)				

5) Unary Operator (Increment / Decrement Operator)



6) Condition / Ternary Operator (?:)

(Condition) ? (True statement) : (false statement);

```
Scanner scn=new Scanner(System.in);
int n1=scn.nextInt();
int n2=scn.nextInt();
int max=(n1>n2)?n1:n2;
System.out.println(max);
```

```
class Solution {
    public int greatestOfThree(int a, int b, int c) {
        // code here
        return (a>b)?(a>c?a:c):(b>c?b:c);
    }
}
```

greatest of
3 numbers

Bitwise Operator (2, 1, ^, <<, >>)

		<u>2</u>	<u>1</u>	<u>1 (same same PUPP same)</u>
1	1	1	1	0
1	0	0	1	1
0	1	0	1	1
0	0	0	0	0

$$5 - 101$$

$$7 - 111$$

$$2 \quad \underline{101} - 5$$

$$1 \quad \underline{111} - 7$$

$$1 \quad \underline{010} - 2$$

① odd or Even (without %)

$$\dots, 2^5 \quad 2^4 \quad 2^3 \quad 2^2 \quad 2^1 \quad \boxed{2^0} \text{ odd}$$

$$\underline{32 \quad 16 \quad 8 \quad 4 \quad 2}$$

$$n \Rightarrow$$

$$(2)$$

$$1 \Rightarrow$$

1	0	0	0	0	1
0	0	0	0	0	1
0	0	0	0	0	1

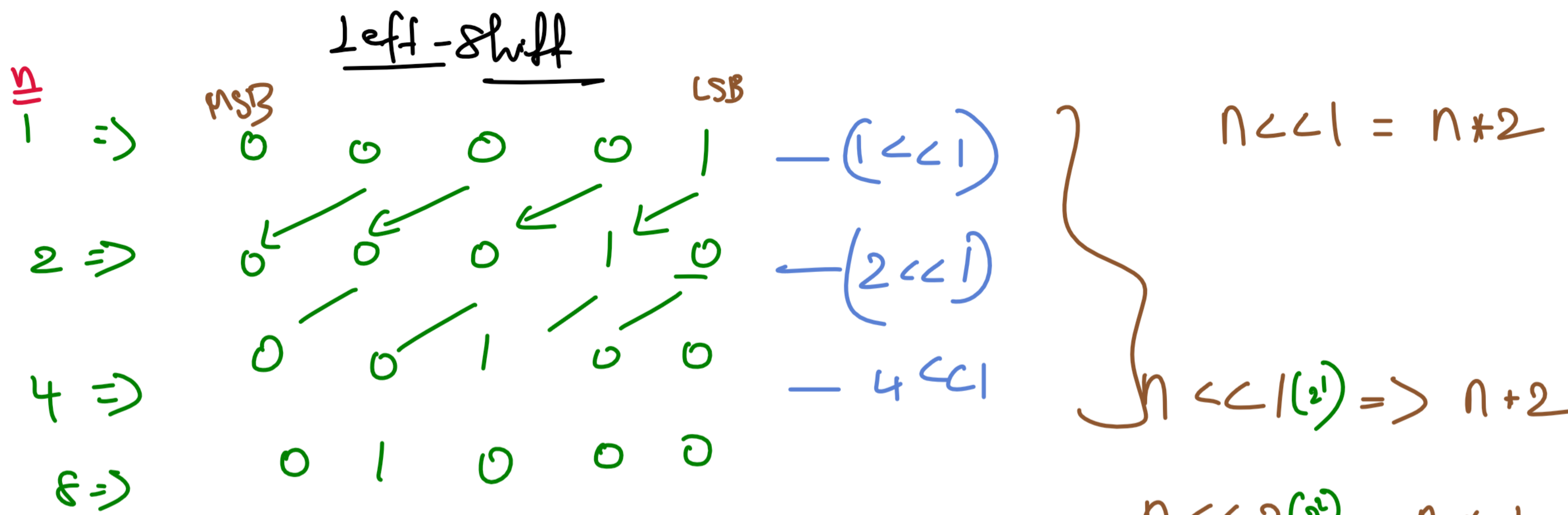
$$(n \& 1) \begin{cases} 1 \text{ (odd)} \\ 0 \text{ (Even)} \end{cases}$$

$$n/2 == 0 \text{ — Even}$$

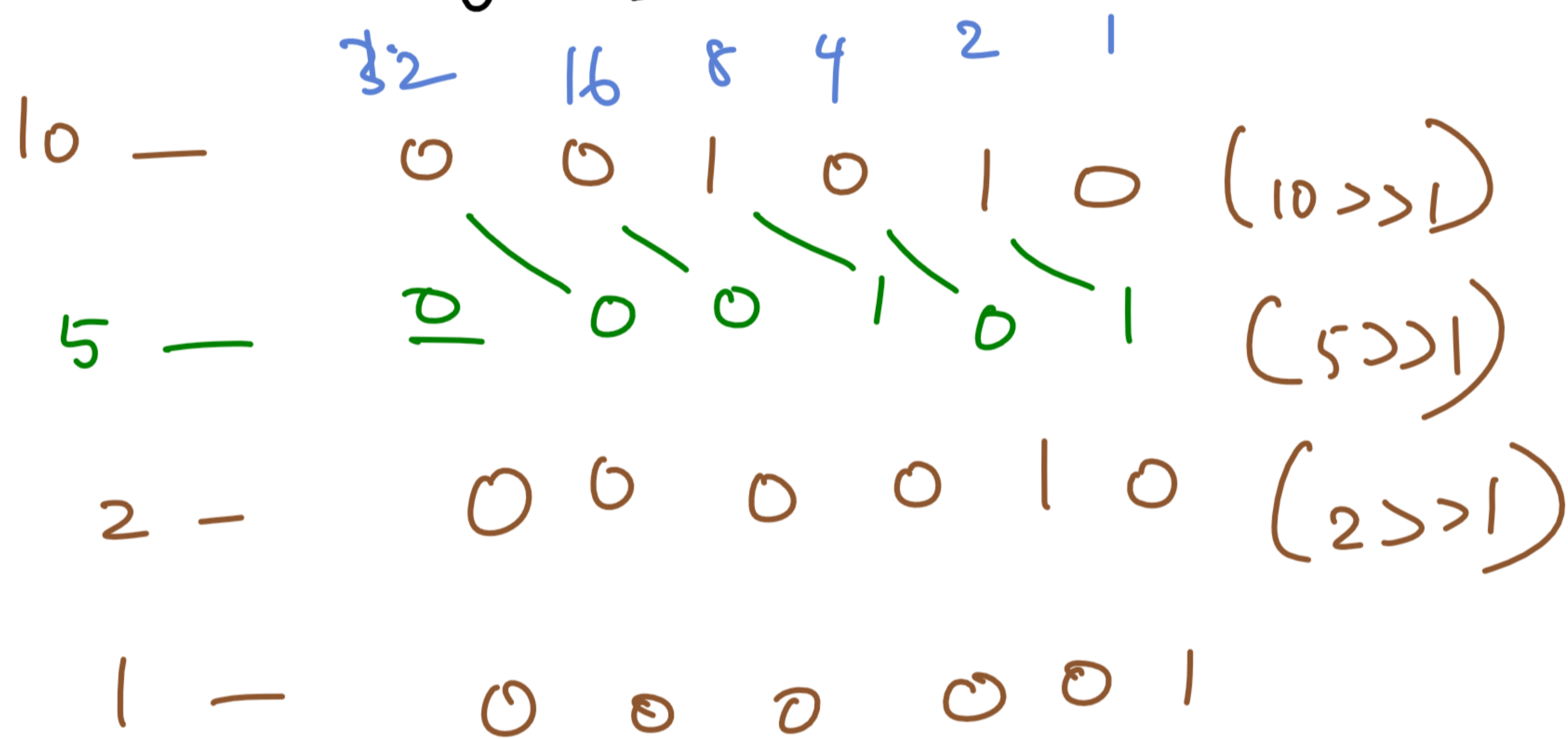
$$== 1 \text{ — odd}$$

$$(n \& 1) == 0 \text{ — Even,}$$

$$== 1 \text{ — odd.}$$



Right shift



$n >> 1 = n/2$

$n >> 2 = n/4$

$n >> 3 = n/8$

```
//Age Eligible
import java.util.*;
public class Main{
    public static void main(String args[]){
        Scanner scn=new Scanner(System.in);
        int age=scn.nextInt();
        System.out.println(age>21?"Eligible":"Not Eligible");
    }
}
```